

Team 7 - E-Waste Battery Extraction

CV-17 Localisation Validation

Experimental Method and Results

Gazebo Ground Truth, Automatic Centroid Extraction and Error Analysis

ROS 2 Jazzy / Gazebo / OpenCV

Validation report

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1 Validation Objective

The aim of the CV-17 validation experiment was to measure how accurately an image-space battery centre could be projected into Gazebo world coordinates using the pixel-to-world geometry completed in Part A. The experiment was designed to isolate geometric projection accuracy from the known detector bounding-box issue.

For each test position, the experiment recorded:

$$(u, v), \quad (X_{pred}, Y_{pred}, Z_{pred}), \quad (X_{GT}, Y_{GT}, Z_{GT})$$

and calculated planar localisation error:

$$e = \sqrt{(X_{pred} - X_{GT})^2 + (Y_{pred} - Y_{GT})^2}. \quad (1)$$

2 Verified Inputs

The validation used the live Gazebo camera calibration:

$$f_x = f_y = 1108.7654113769531, \quad c_x = 640, \quad c_y = 480,$$

with 1280 by 960 images and zero simulated distortion. The verified camera transform was:

$$(C_x, C_y, C_z) = (0.0, 0.60, 1.90) \text{ m}$$

with the ROS optical frame rotated into the world frame through the permanent TF `world <- top_camera_optical_frame`.

```
===== BATTERY_1 GROUND TRUTH =====
Requesting state for world [ewaste_room]...
Model: [28]
- Name: battery_1
- Pose [ XYZ (m) ] [ RPY (rad) ]:
  [-0.149998 0.619999 0.800000]
  [-0.000004 -0.000014 0.000173]

===== LOCALISATION TF =====
[INFO] [1788612286.428669975] [tf2_echo]: Waiting for transform world -> top_camera_optical_frame: Invalid frame ID "world" passed to canTransform argument
target_frame - frame does not exist
At time 0.0
- Translation: [0.000, 0.600, 1.900]
- Rotation: in Quaternion (xyzw) [-0.707, 0.707, -0.000, 0.000]
- Rotation: in RPY (radian) [-3.142, -0.000, -1.571]
- Rotation: in RPY (degree) [-180.000, -0.000, -90.000]
- Matrix:
  -0.000 -1.000 0.000 0.000
  -1.000 0.000 -0.000 0.600
  0.000 -0.000 -1.000 1.900
  0.000 0.000 0.000 1.000
At time 0.0
- Translation: [0.000, 0.600, 1.900]
- Rotation: in Quaternion (xyzw) [-0.707, 0.707, -0.000, 0.000]
- Rotation: in RPY (radian) [-3.142, -0.000, -1.571]
- Rotation: in RPY (degree) [-180.000, -0.000, -90.000]
- Matrix:
  -0.000 -1.000 0.000 0.000
  -1.000 0.000 -0.000 0.600
  0.000 -0.000 -1.000 1.900
  0.000 0.000 0.000 1.000
```

Figure 1: Live ground truth and verified camera TF used by the validation pipeline.

3 Centroid Method Selection

3.1 YOLO Output Was Not Used for Geometry Accuracy

The live detector produced a high-confidence battery-class result, but the reported bounding box covered a very large part of the image. Its centroid therefore did not correspond to `battery_1`. Using that centroid would have measured detector error rather than pixel-to-world geometry.

```
ad@DESKTOP-OT45K60:~/team7_project_bundle/team7_perception_ws$ timeout 10 ros2 topic echo /phone_detections --once
header:
  stamp:
    sec: 696
    nanosec: 0
  frame_id: overhead_camera/mount/overhead_camera
detections:
- header:
  stamp:
    sec: 696
    nanosec: 0
  frame_id: overhead_camera/mount/overhead_camera
  class_id: battery
  score: 0.9073662757873535
  bbox_x1: 880
  bbox_y1: 19
  bbox_x2: 1278
  bbox_y2: 911
  mar_center_x: 1078.5
  mar_center_y: 464.49993896484375
  mar_width: 396.99993896484375
  mar_height: 890.9998779296875
  angle_deg: 90.0
  world_x: 0.015796911411798856
  world_y: 0.15310204041736597
  world_z: 0.77
  world_valid: true
---
```

Figure 2: Detector output that was deliberately excluded from the formal geometry benchmark.

3.2 Manual-Centroid Pilot

A manual click tool was first used as an independent check. It successfully converted the selected pixel into a world coordinate and automatically compared it with live Gazebo ground truth. However, different manual clicks on the same battery body varied by several pixels and produced noticeably different millimetre errors.



Figure 3: Manual-centroid pilot test. Human click placement was a measurable source of uncertainty.

3.3 Automatic Green Segmentation

The final experiment used a deterministic, YOLO-independent centroid method. The validator:

1. subscribes to `/top_camera/image` and `/top_camera/camera_info`;
2. converts the image to HSV;
3. thresholds the green battery colour;
4. applies morphological opening and closing;
5. selects the largest green contour above a minimum area;
6. fits a minimum-area rectangle;
7. uses the rectangle centre as (u, v) ;
8. predicts the world point with `pixel_to_world_tf`;
9. queries `battery_1` ground truth from Gazebo;
10. saves the result to CSV and writes an evidence image.

The contour-moment centroid was also recorded for diagnostic comparison with the MAR centre.

4 Experimental Design

A fixed battery-centre plane of $Z_p = 0.800$ m was used for all tests. Five positions were selected to cover both the interior and corners of the validation region.

Table 1: Validation positions used in the formal experiment.

Position	GT X (m)	GT Y (m)	GT Z (m)
P1	-0.149998	0.619999	0.800000
P2	-0.350000	0.849993	0.800000
P3	0.100002	0.850006	0.800000
P4	0.100003	0.500019	0.800000
P5	-0.349999	0.499996	0.800000

The battery was moved with Gazebo’s `/world/ewaste_room/set_pose` service and allowed to settle before each measurement. Ground truth was read from the live world state after settling.

5 Formal Measurement Results

Table 2: World-coordinate prediction and planar localisation error.

Pos.	Pixel u	Pixel v	Pred. X	Pred. Y	dX (mm)	dY (mm)
P1	618.500	632.000	-0.150798	0.621330	-0.80	+1.33
P2	385.961	833.308	-0.350515	0.852030	-0.51	+2.04
P3	385.800	378.900	0.100301	0.852190	+0.30	+2.18
P4	741.250	377.750	0.101442	0.499550	+1.44	-0.47
P5	741.661	833.751	-0.350954	0.499143	-0.96	-0.85

Table 3: Planar error at each validation position.

Position	XY error (mm)
P1	1.55
P2	2.10
P3	2.20
P4	1.51
P5	1.28

6 Aggregate Statistics

The formal CSV was processed directly to produce the final summary:

Mean XY error = 1.731 mm,
 RMSE = 1.767 mm,
 Minimum error = 1.281 mm,
 Maximum error = 2.205 mm,
 Mean dX bias = -0.107 mm,
 Mean dY bias = +0.846 mm.

```

===== CV-17 PART B FINAL VALIDATION =====
Formal measurements: 5

Pos      GT X      GT Y      Pred X      Pred Y      dX mm      dY mm      Error mm
P1    -0.149998  0.619999 -0.150798  0.621330    -0.80      1.33      1.55
P2    -0.350000  0.849993 -0.350515  0.852030    -0.51      2.04      2.10
P3     0.100002  0.850006  0.100301  0.852190     0.30      2.18      2.20
P4     0.100003  0.500019  0.101442  0.499550     1.44     -0.47      1.51
P5    -0.349999  0.499996 -0.350954  0.499143    -0.96     -0.85      1.28

===== AGGREGATE STATISTICS =====
Mean XY error : 1.731 mm
RMSE          : 1.767 mm
Minimum error : 1.281 mm
Maximum error : 2.205 mm
Mean dX bias  : -0.107 mm
Mean dY bias  : +0.846 mm

Summary saved to: /home/ad/team7_project_bundle/team7_perception_ws/cv17_validation_summary.csv

```

Figure 4: Terminal output of the final five-position validation table and aggregate statistics.

7 Result Visualisation

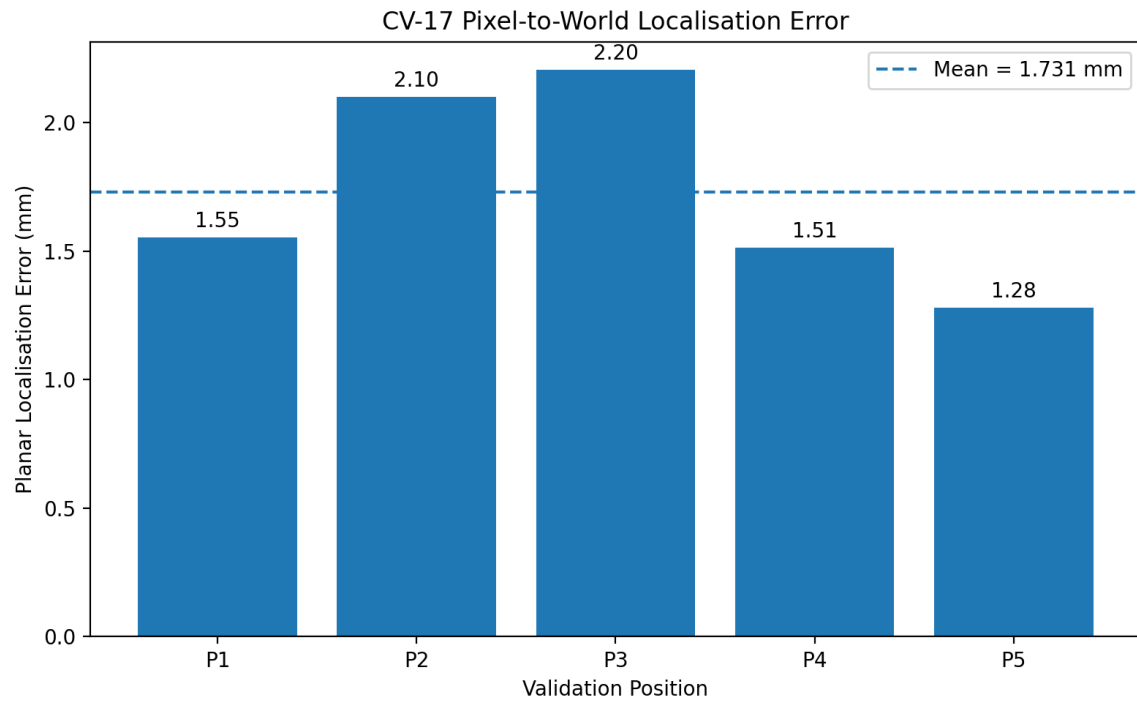


Figure 5: Planar localisation error at P1-P5. Every test remained below 2.21 mm.

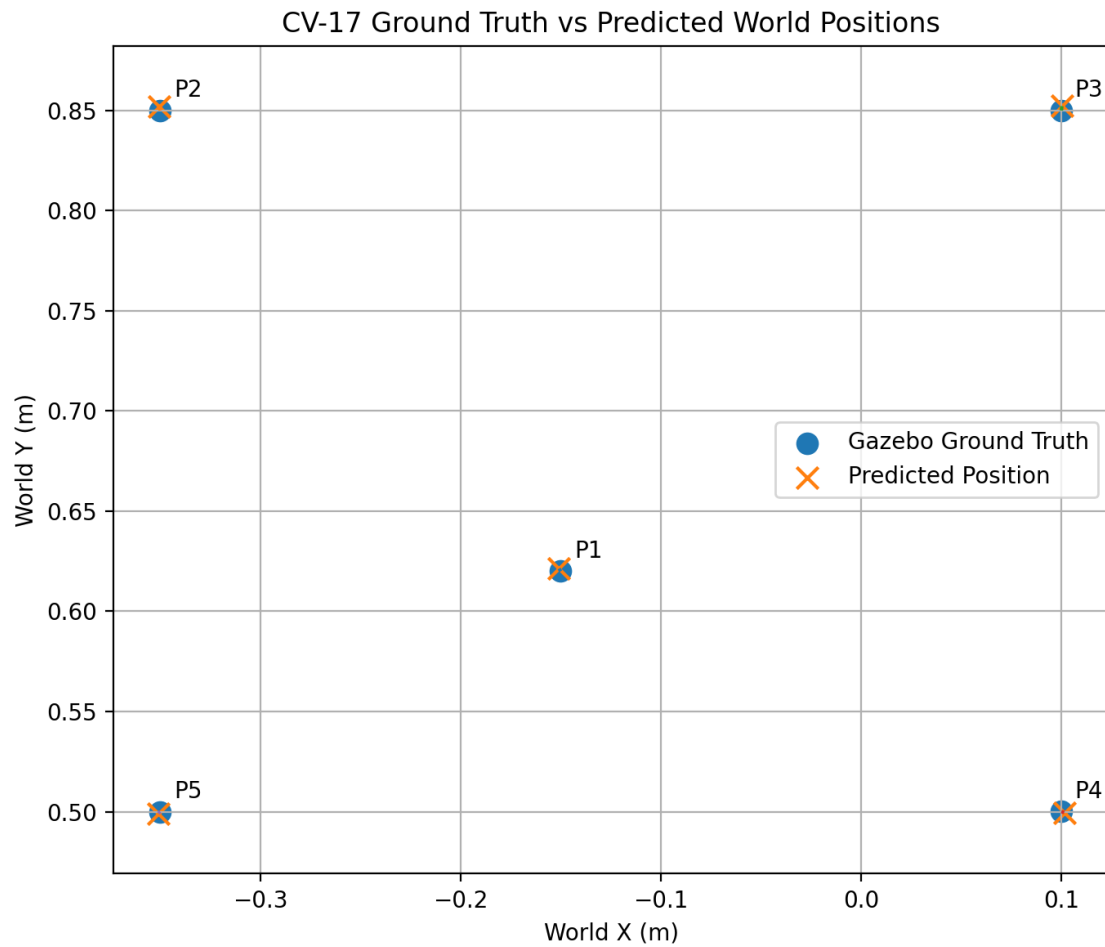


Figure 6: Gazebo ground-truth positions and predicted world positions. Predicted and ground-truth markers nearly overlap throughout the test region.

8 Repeatability and Robustness

The final method removed the main human source of uncertainty observed in the manual-click pilot. The automatic MAR centre and contour centroid were consistently close to each other, indicating that the segmented green region was stable enough for the controlled validation object.

A practical runtime issue occurred when the image topic temporarily stopped delivering frames within the original 10-second validator window even though the topic remained registered and `camera_info` continued at about 1 Hz. The waiting period was increased to 180 seconds, after which the same validation script completed without changing the geometry or simulation state.

```
ad@DESKTOP-OT45K60:~/team7_project_bundle/team7_perception_ws$ timeout 6 ros2 topic hz /top_camera/image
WARNING: topic [/top_camera/image] does not appear to be published yet
ad@DESKTOP-OT45K60:~/team7_project_bundle/team7_perception_ws$ timeout 6 ros2 topic hz /top_camera/camera_info
average rate: 1.030
  min: 0.895s max: 1.047s std dev: 0.07574s window: 2
ad@DESKTOP-OT45K60:~/team7_project_bundle/team7_perception_ws$ ros2 topic list | grep top_camera
/top_camera/camera_info
/top_camera/image
ad@DESKTOP-OT45K60:~/team7_project_bundle/team7_perception_ws$
```

Figure 7: Camera-stream timeout observed during repeated Gazebo/WSL testing.

9 Interpretation and Limitations

The experiment supports the conclusion that the implemented pixel-to-world geometry is accurate to a few millimetres across the tested region. The maximum observed error was 2.205 mm and the mean error was 1.731 mm. The small signed biases also indicate that no large systematic axis reversal or scale error remained after the TF integration.

The result is specifically a **planar geometry validation**. The target height was fixed at the known settled battery-centre height, so Z was not independently estimated. The formal metric is therefore XY localisation error.

The automatic green segmentation was intentionally used only as an independent experimental centroid source. It is not proposed as a production replacement for the ONNX detector. End-to-end detector-to-world accuracy should be measured separately after the detector's current battery bounding-box issue is corrected.

10 Conclusion

CV-17 Part B validated the Part A localisation geometry against live Gazebo ground truth at five distinct world positions. The final experiment was reproducible, independent of the faulty YOLO centroid, and generated both numerical and visual evidence. The final mean planar error was 1.731 mm, RMSE was 1.767 mm, and the maximum error was 2.205 mm.

These results provide quantitative evidence that the camera intrinsics, TF frame convention, quaternion ray transformation and ray-plane intersection are mutually consistent in the supplied Team 7 Gazebo environment.

A Reproduction Commands

A.1 Move a Validation Object

```
gz service -s "/world/ewaste_room/set_pose" \
  --reqtype gz.msgs.Pose \
  --reptype gz.msgs.Boolean \
  --timeout 3000 \
  --req 'name: "battery_1" position {x: ... y: ... z: 0.81} \
        orientation {x: 0 y: 0 z: 0 w: 1}'
```

A.2 Check Ground Truth

```
gz model -m battery_1 -p
```

A.3 Run Automatic Validation

```
python3 cv17_validate_green.py
```

The formal run writes `cv17_validation_auto_results.csv`, `cv17_validation_summary.csv`, and the annotated images/plots under `cv17_validation_evidence/`.